

Mohammed Azharudeen Farook Deen

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📁 Portfolio Website in fmdazhar 🌐 fmdazhar

About

Robotics Engineer with a focus on **deep robot learning** and **optimal control** to build intelligent and safe robots.

Skills

Programming: Python, C++, MATLAB

Machine Learning & AI: PyTorch, JAX, TensorFlow, Deep Learning (CNNs, RNNs, Transformers), Reinforcement Learning (PPO, SAC), Imitation Learning (BC, DAgger), Robot Foundation Models (NVIDIA GROOT, pi-zero)

Robotics & Simulation: ROS, ROS2, Isaac Sim, Isaac Gym, Mujoco, NVIDIA Warp

Optimization & Control: Model Predictive Control (MPC), Quadratic/Nonlinear Programming (QP, SQP)

Development Tools & Systems: Git, Docker, Linux

Languages: English (Business Fluency), German (Beginner), Tamil (Native)

Work Experience

Intern

Fraunhofer IPA

Stuttgart, Germany

July 2024 – Mar 2025

- Developed a **custom simulation environment in MuJoCo and ROS**, supporting both state- and vision-based inputs for a peg-in-hole assembly task with different difficulty presets.
- Implemented and tuned various controller modules; Integrated teleoperation modules for keyboard and spacemouse control for **Learning from Demonstrations (LfD)** and **Human-in-the-Loop RL**.
- Built a real robot (UR5e + Robotiq hand-e Gripper) interface with **Flask-ROS server** for command execution with **actor-learner asynchronous nodes**.
- Adapted both the simulation and real environment to work with real-time online Reinforcement Learning (SAC) and Imitation Learning (BC, DAgger) pipelines in both **Pytorch**, and **JAX** and benchmarked them using simulation.

Master Thesis - Robot Learning of Quadrupedal Locomotion on Deformable Terrain

RWTH Institute for Data Science in Mechanical Engineering (DSME) and RWTH Institute of Geomechanics and Underground Technology (GUT)

Aachen, Germany

Sep 2023 – Mar 2024

- Integrated a particle-based contact model to simulate the complex behaviors of granular materials in **NVIDIA Isaac Sim**, developed an adaptable and robust controller using Reinforcement Learning (PPO) in PyTorch for quadruped (Unitree A1) to follow a unified policy in diverse natural environments and evaluated the performance with sim-to-real additions.
- Preliminary work accepted at the 1st German Robotics Conference (GRC2025), and a related paper is being finalized for publication.

Student Assistant

RWTH Institute for Automatic Control (RWTH IRT)

Aachen, Germany

May 2023 – June 2024

- Extended codebase of a unified C++ interface for various **quadratic programming solvers (QP)** to **nonlinear programming (NLP) solvers** and included coupled tools for MPC, NMPC, SQP, auto differentiation, Hessian regularization methods, etc with appropriate test cases.
- Involved in the Fernbin project, where my tasks included developing simple simulation for testing control systems in Python and **ROS2** for ships that react dynamically to the surroundings.

Research Project

Automated and Connected Driving Challenges — RWTH Institute for Automotive Engineering (ika)

Aachen, Germany

Apr 2023 – Aug 2023

- Enhanced trajectory planning in autonomous vehicles using **Model Predictive Control (MPC)**.

Teaching Assistant

RWTH Institute of Geomechanics and Underground Technology (RWTH GUT)

Aachen, Germany

Apr 2023 – Sep 2023

- Implemented programming exercises in Python/ROS, conducted tutorial sessions for “Introduction to Robotics” course.

Research Assistant

Microscale Transport Laboratory, Indian Institute of Information Technology (IIITDM))

Kancheepuram, India

Aug 2019 – Mar 2021

- Studied predominant microscale phenomena subjected to acoustic fields and published results in various reputed journals.

Education

RWTH Aachen University
*Master of Science in **Robotic Systems Engineering***

Oct 2021 – Dec 2024

Shiv Nadar University
*Bachelor of Technology with major in **Mechanical Engineering**(Specilization: Computational Techniques) and minor in Electronics and Communication Engineering*

Aug 2015 - May 2019

Journal Publications and Accepted Conferences

Rapid Quadrupedal Locomotion on Deformable Terrain - 1st German Robotics Conference

March 2025

Mohammed Azharudeen Farook Deen, Omer Kemal Adak, Raul Fuentes
<https://www.robotics-institute-germany.de/conference/> 

Theory of nonlinear acoustic forces acting on inhomogeneous fluids. Journal of Fluid Mechanics. [IF(2022)-3.7]

April 2022

Rajendran VK, Jayakumar S, Azharudeen M, Subramani K
<https://doi.org/10.1017/jfm.2022.257> 

Heat transfer mechanism driven by acoustic body force under acoustic fields. Physical Review Fluids. [IF(2021)-2.5]

July, 2021

Azharudeen M, Kumar V, Pothuri C, Subramani K

A Novel Heat Transfer Mechanism Using Acoustic Waves - Presented at International Heat and Mass Transfer Conference, IIT Roorke

Dec 2019

Azharudeen M, Pothuri C, Subramani K

<https://doi.org/10.1103/PhysRevFluids.6.073501> 

Rapid mixing in microchannel using standing bulk acoustic waves. Physics of Fluids. [IF(2020)-3.5]

Dec 2019

Azharudeen M, Pothuri C, Subramani K

<https://doi.org/10.1017/jfm.2022.257> 

Honors & Awards

Full Academic Scholarship, 8 Semesters – Shiv Nadar University

Aug 2015

Issued by Shiv Nadar University

Completed my engineering program with 100% scholarship across all 8 semesters.